ML08 Select, filter and calculate new columns

Week 3, Tables and statistics | Session 8 of 24 | Machine Learning | New library: pandas

Today's question. Which item earned the most money in the practice shop?

Objective. By the end: you can pick a column, keep rows that meet a condition, add a revenue column and total it by item.

> **WHERE THIS IS USED**
> A shopkeeper asks which item earned the most this week. Answering needs a filter and a new column: units times price. What would you ask of the sales of your own shop?

Before this session: ML07 Tables and reading a CSV file. If you cannot explain the last example, use its error card with the instructor first.

Words for today

Word	What it means	Think of it like this
select a column	Name the column in square brackets to look only at it.	Reading only the price line of the register.
filter	Keep only the rows where a comparison is True.	Picking out the sales of 5 units or more.
new column	A column calculated from other columns, added to the table.	Multiplying units by price on every line.
sum	All the values added together.	Adding up the day's takings.
group	Gather rows with the same label before calculating.	Sorting the sales into three piles by item.
revenue	Money received from sales: units times price. It is not profit.	The takings before paying for stock.

Block A. Select and filter

Read the practice shop and keep the large sales.

Predict. Predict how many of the 60 sales have 5 units or more. Write your guess on paper first.

Your guess, written before you run it

```
import pandas as pd
shop = pd.read_csv("camp_shop.csv")
units = shop["units"]
big = shop[shop["units"] >= 5]
print(len(big))
print(units.sum())
```

Run it, then compare. The cell prints 30 and 277. shop["units"] selects one column. The comparison inside the outer brackets gives True or False for every row, and only the True rows stay. That filter keeps 30 sales. All 60 sales together sold 277 units.

Investigate.

1. Is the filter deleting the other 30 rows?
2. Why can the sum of units not tell us the money earned?

Modify. Filter the Juice sales with 5 units or more and count them.

Block B. A revenue column and totals by item

Multiply units by price on every row, then add the money for each item.

Predict. Predict which item earns the most money. Is it the one with the most units? Write your guess on paper first.

Your guess, written before you run it

```
shop["revenue_ugx"] = shop["units"] * shop["unit_price_ugx"]
total_revenue = shop["revenue_ugx"].sum()
by_item = shop.groupby("item")["revenue_ugx"].sum()
print(total_revenue)
print(by_item)
```

Run it, then compare. The total is 550000 shillings. Book earns 297000, Juice 150000 and Pen 103000. Pen sold the most units, 103, but Book earns the most money because each Book costs 3000. Assigning to a new column name creates the column, and the groupby line adds the revenue of each item separately.

Investigate.

1. Pen sold the most units. Why is it last for revenue?
2. Do the three item totals add up to the overall total?

Modify. Group the units column by item instead and print the totals.

Make

Keep the rows with units of at least 3. Calculate the total units and the mean units per row. Say why these two numbers are different quantities.

Save your work as ML08. Keep the worked example and put your own attempt below it. Write two sentences: what you changed and how you checked it.

Error card

You see	It means	Fix
KeyError: 'unit'	The column name is spelled differently from the header.	Print the headers with list(shop.columns) and copy the exact name.
The filter returns an empty table	The comparison matches no rows, often because of a typo in a text value.	Check the spelling and capital letters of the value you compare with.

Exit ticket

A filter keeps the rows I ask for, and a grouped total shows who earns the money.

AI use follows AI_USE_POLICY.pdf. In weeks 1 to 6 no AI-generated solutions are used for exercises, labs or build tasks. Practise for 20 to 30 minutes between classes by repeating the Make task with one value changed. If you are stuck for 30 minutes, save the error and bring it to class. A file saved on your laptop has not been shared until you use the route in START_HERE.pdf.